

Quadratic Equations (Set B) — MCQ Worksheet

Mathematics • Chapter 4 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Which is a quadratic equation?

(A) $x + 2 = 0$

(C) $x^3 + x = 0$

(B) $x^2 + 3x - 1 = 0$

(D) $x + 1/x = 0$

Q2. Roots of $x^2 - 3x - 10 = 0$ are:

(A) 5, -2

(C) 5, 2

(B) -5, 2

(D) -5, -2

Q3. Discriminant of $3x^2 - 5x + 2 = 0$ is:

(A) 1

(C) 25

(B) -1

(D) 49

Q4. Nature of roots of $x^2 + 6x + 9 = 0$ is:

(A) Real and distinct

(C) Not real

(B) Real and equal

(D) Imaginary distinct

Q5. If roots of $ax^2 + bx + c = 0$ are equal, then:

(A) $b^2 = 4ac$

(C) $b^2 < 4ac$

(B) $b^2 > 4ac$

(D) $b^2 = 0$

Q6. For $x^2 - 7x + 12 = 0$, sum of roots is:

(A) 7

(C) 12

(B) -7

(D) -12

Q7. Product of roots of $x^2 - 4x + 3 = 0$ is:

(A) 3

(C) 4

(B) -3

(D) -4

Q8. If roots of $x^2 - kx + 4 = 0$ are equal, k is:

(A) ± 2

(C) ± 6

(B) ± 4

(D) ± 8

Q9. A quadratic equation with roots 3 and -2 is:

(A) $x^2 - x - 6 = 0$

(C) $x^2 + x + 6 = 0$

(B) $x^2 + x - 6 = 0$

(D) $x^2 - 5x + 6 = 0$

Q10. Number of real roots of $(x - 3)^2 + 4 = 0$ is:

(A) 0

(C) 2

(B) 1

(D) Infinitely many

Q11. If one root of $2x^2 + kx + 1 = 0$ is $-1/2$, then k =

(A) 3

(C) 2

(B) -3

(D) -2

Q12. Roots of $x^2 - 2x - 8 = 0$ are:

(A) 4, -2

(C) 4, 2

(B) -4, 2

(D) -4, -2

Q13. For $x^2 + 4x + 4 = 0$, the discriminant is:

(A) 0

(C) -16

(B) 16

(D) 32

Q14. Sum and product of roots of $3x^2 - 6x + 4 = 0$ are:

(A) 2 and $4/3$

(C) 2 and $-4/3$

(B) -2 and $4/3$

(D) 6 and 4

Q15. For the equation $x^2 + px + 1 = 0$ to have no real roots, p must satisfy:

(A) $-2 < p < 2$

(B) $p > 2$ or $p < -2$

(C) $p = \pm 2$

(D) $p \geq 2$

Q16. If the difference of roots of $x^2 - 13x + k = 0$ is 7, then $k =$

(A) 30

(B) 31

(C) 32

(D) 40

Q17. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less. The speed of train is:

(A) 30 km/h

(B) 40 km/h

(C) 45 km/h

(D) 50 km/h

Q18. Find two consecutive positive integers, sum of whose squares is 365.

(A) 12, 13

(B) 13, 14

(C) 14, 15

(D) 10, 11

Q19. The sum of the reciprocals of two numbers is $\frac{5}{8}$, and their sum is 5. The numbers are:

(A) 1 and 4

(B) 2 and 3

(C) 2.5 each

(D) 1.5 and 3.5

Q20. A rectangular park has length 5 m more than its breadth, and area 50 m^2 . Length of park is:

(A) 10 m

(B) 12 m

(C) 8 m

(D) 15 m

Q21. If the roots of equation $(b - c)x^2 + (c - a)x + (a - b) = 0$ are equal, then:

(A) $a + b = 2c$

(B) $b + c = 2a$

(C) $a + c = 2b$

(D) $a = b = c$

Q22. If α, β are roots of $x^2 + px + q = 0$, then $\alpha^2 + \beta^2 =$

(A) $p^2 - 2q$

(B) $p^2 + 2q$

(C) $p^2 - q$

(D) $2p^2 - q$

Q23. Find k so that the equation $2x^2 + kx + 3 = 0$ has real and equal roots:

(A) $k = \pm 2\sqrt{6}$

(B) $k = \pm \sqrt{6}$

(C) $k = \pm 4$

(D) $k = \pm 6$

Q24. The sum of squares of two consecutive odd positive integers is 290. The integers are:

(A) 11, 13

(B) 9, 11

(C) 13, 15

(D) 7, 9

Q25. A two-digit number is such that the product of its digits is 14. If 45 is added to the number, the digits are reversed. The number is:

(A) 27

(B) 72

(C) 29

(D) 92